

How to build pricing engines... — Matrix Profiles for Pattern Recognition in Financial Time Series

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Couldn't resist to starting on another topic that I have on my to-do list: pattern recognition.

The screenshot below shows some functionality I implemented already, that uses "Matrix Profiles" as a pattern-recognition engine for financial data.

Data is the daily prices of a major listed Swiss insurance company.

On the left, you see the core outputs of the Matrix Profile for a chosen window size (here, 10 bars). For every 10-point window in the return series, the add-in computes a profile value (a distance) and a nearest-neighbor index. Low profile values highlight local patterns that recur elsewhere in the series; high values mark discords, i.e. unusual sequences of returns that have no close analogue in the past. The nearest-neighbor index tells us where in the time series that closest match occurs, so we can pull out concrete "pattern pairs" instead of abstract statistics.

In the middle section, those pairs are turned into actual windows of data. The sheet shows the selected windows side by side, both in raw form and standardized form. This makes it visually obvious that we're not just correlating levels, but comparing shapes: 10-bar paths that look strikingly similar once you remove mean and volatility. It's a way of asking, in a very literal sense: "when did the market trade through a similar micro-path?".

The charts at the bottom and right connect these pattern measures back to the time series. One chart overlays profile values with cumulative returns, so you can see how motif-rich and anomaly-rich periods line up with the market's trajectory.

The other charts show the most similar 10-bar patterns in raw and standardized space, demonstrating how the Matrix Profile can surface regime signatures and recurring micro-structures that are hard to spot with traditional tools. Together, this turns abstract similarity scores into a tangible pattern-recognition overlay on price and return series.

My goal is to bring some state-of-the-art time-series motif and anomaly detection directly into Excel with ApaLibNET - not as a black-box ML model, but as a transparent, inspectable toolkit for discovering patterns we can actually see, compare, and reason about, without being a programmer or becoming one.

The math methods available in pattern recognition can become mind-boggling rather quickly; I will restrict myself to the rudimentary methods on purpose.

Source: https://www.linkedin.com/posts/andreassteiner_coulnt-resist-to-starting-on-another-topic-share-7481611775717654528-HPLK/